(4) $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$

(5) Ak question C.R

Date:

$$\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$$

$$\frac{4z^2}{z^2+2z+1}$$

$$\frac{z^2(4)}{z^2\left(1-\frac{2}{z}+\frac{1}{z^2}\right)}$$

$$\frac{z^2(4)}{z^2\left(1-\frac{2}{z}+\frac{1}{z}\right)}$$

$$\lim_{z \to \infty} \frac{4}{1-\frac{2}{\infty}+\frac{1}{\infty}}$$

$$\frac{4}{1-0+0} = 4 \quad \text{R.H.S}$$

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Date:

$$\lim_{2+i} \; z^2 + 2z + 9 - 8 + 5i$$

$$(2+i)^2 + 2(2+i) - 8 + 5i$$

$$4 + i^2 + 2i + 4 + 2i - 8 + 5i$$

$$4 - 1 + 4i + 4 + 2i - 8 + 5i$$

$$4 - 1 + 4 - 8 + 4i + 2i + 5i$$

$$-1 + 11i$$

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$f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

$x = \frac{z+\bar{z}}{2}$ $\quad y = \frac{z-\bar{z}}{2i}$

$$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right) + i\left(2\left(\frac{z+\bar{z}}{2}\right) - 2\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right)\right)$$

$$= \frac{z^2+\bar{z}^2+2z\bar{z}}{4} - \left(\frac{z^2+\bar{z}^2-2z\bar{z}}{4}\right) - \frac{2(z-\bar{z})}{2i}$$

$$\frac{4z\bar{z}}{4} - \frac{2(z-\bar{z})}{2i} + i\left(\frac{z+\bar{z}-(z^2-\bar{z}^2)}{i}\right)$$

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$$|z|^2 = z\bar{z}$$

$z = x + iy$

$|z|^2 = \left(\sqrt{x^2 + y^2}\right)^2$

$|z|^2 = x^2 + y^2$

$z = x + iy$

$\bar{z} = x - iy$

$z\bar{z} = x^2 - i^2 y^2$

$z\bar{z} = x^2 + y^2$

$z\bar{z} = x^2 + y^2$

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$$(1+\sqrt{3}i)^4$$

$1+\sqrt{3}i =$

$$r = \sqrt{1+3} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}/1)$$

$$\pi/3$$

$$= 2e^{i(\pi/3)}$$

$$(1+\sqrt{3}i)^4 = \left(2(\cos\pi/3 + i\sin\pi/3)\right)^4$$

$$2^4\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right)$$

$$2^4 e^{i4\pi/3}$$

$$16e^{i4\pi/3}$$

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Real Limits of Multivariable functions

Example: Show that the function $u(x,y) = \frac{xy}{x^2+y^2}$ does not have a limit as (x,y) approaches $(0,0)$.

Solution: Along $y = x$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+x^2}$$

along $y = x$

$$= \lim_{x\to 0} \frac{x^2}{2x^2}$$

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Tools View Help

Real Limits of Multivariable functions

Example: If $u(x,y) = \frac{2x^3}{(x^2+y^2)}$ then show that

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0 .$$

Solution:

Find $\delta > 0$ such that

$|u(x,y)-0| \le 2h < \varepsilon$ whenever $0 < \underbrace{\sqrt{(x-0)^2+(y-0)^2}}_{h} < \delta$

or $|u(x,y)-0| \le 2h < \varepsilon$ • $h < \delta$

choose $\boxed{\delta = \frac{\varepsilon}{2}} \Rightarrow h < \frac{\varepsilon}{2} \Rightarrow \boxed{2h < \varepsilon} \Rightarrow |u-0| < \varepsilon$

VU

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Continuity of $u(x, y)$

nple: Check whether the following function is nuous at (0,0)

$$u(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

tion:

$$\lim_{(x,y)\to(0,0)} \frac{x^3}{x^2+y^2} = 0$$

$u(0,0) = 0$

$u(x,y)$ is cont. at $(0,0)$

15:02

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inuity of Complex Functions

: Show that the function $f(z) = \bar{z}$ is continuous.

:

$\lim_{z \to z_0} \bar{z} = \bar{z}_0$ (from previous discussion)

$f(z_0) = \bar{z}_0$

$\Rightarrow$ $f(z)$ is Cont. at z_0

$\Rightarrow$ $f(z)$ is Cont. at each Complex number:

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8:07 4G 81%
You
Yesterday, 6:44 pm

(ix) $\lim_{z \to (i+1)} (z^2 - 2z + 1)$

(x) magnitude of $\exp(-2+3i)$

(xi) $\frac{1}{re^{i\theta}}$ The $U_r = ?$

U_r tha
U_y tha
U_x tha kuch clear nai tha.

(xii) Close disk ki mathematical form puchi hui thi.

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$$|z^3 + 3z^2 - 2z + 1|$$

$$\leq |z^3| + |3z^2| + |-2z| + 1$$

$$\because |z^3| = |z|^3 \quad |z_1 z_2| = |z_1||z_2|$$

$$|z|^3 + 3|z|^2 + 2|z| + 1$$

$$< z^3 + 3z^2 + 2z + 1$$ upper bound

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Date

$$f(z) = (1+z^2)^4$$

$$\cdot z^2$$

$$= 4(1+z^2)^{4-1} \cdot 2z$$

$$2z$$

$$= 4(1+z^2)^3$$

OR Chain Rule

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$+i)^3$

$f(z) = x^2 - y^2 - 2y + i(2x - 2xy)$

$x = \frac{z+\bar{z}}{2} \qquad y = \frac{z-\bar{z}}{2i}$

$\left(\frac{z+\bar{z}}{2}\right)^2 - \left(\frac{z-\bar{z}}{2i}\right)^2 - 2\left(\frac{z-\bar{z}}{2i}\right)$

$+ i\left(2\left(\frac{z+\bar{z}}{2}\right) - 2\left(\frac{z+\bar{z}}{2}\right)\left(\frac{z-\bar{z}}{2i}\right)\right.$



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Q1 Re z and Im z

$z = x + iy$ $\quad \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$2x = z + \bar{z}$

$x = \frac{z + \bar{z}}{2}$

Im z $\quad z - \bar{z} = x + iy - x$

$z - \bar{z} = +2iy$

$y = \frac{z - \bar{z}}{2i}$

$Re z = \frac{z + \bar{z}}{2}$ $\quad Im z =$

Q NO 2

Quadratic Formula

mple: Find roots of the following equation

$$z^2 + (2+i)z + \frac{3}{4} = 0$$

$a=1,\ b=2+i,\ c=3/4$

$$\frac{-(2+i) \pm [(2+i)^2 - 4(1)(3/4)]^{1/2}}{2(1)}$$

$$\frac{-(2+i) \pm [4+i^2+2(2i) - 3]^{1/2}}{2}$$

$$\frac{-(2+i) \pm [4-1+4i-3]^{1/2}}{2}$$

$$\frac{-(2+i) + [4i]^{1/2}}{2}$$

Here $[4i]^{1/2} = \sqrt{2} + i\sqrt{2}$, $-\sqrt{2} - i\sqrt{2}$

Roots of Complex Numbers (Part II)

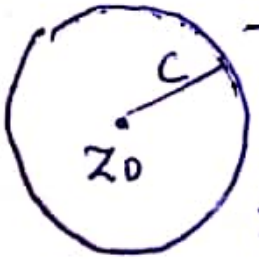
So, $\frac{-(2+i) \pm (\sqrt{2}+i\sqrt{2})(2-i\sqrt{2})}{2}$

$\frac{-(2+i) + (\sqrt{2}+i\sqrt{2})}{2}$, $\frac{(-2+i) \pm (-2\sqrt{2} - \sqrt{2}i)}{2}$.

END

Roots of Unity
Relation between roots of complex numbers and roots of unity
- Quadratic formula

A circle in plane is a set of points whose distance from a fixed point, say z_0 is constant



The point z_0 is called centre, the constant is called radius

* Equation of a circle of radius ε cent z_0 is $|z - z_0| = \varepsilon$

A closed disc with centre z_0 and ra $|z - z_0| \leq \varepsilon$

in the Complex Pla

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given by

$$z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right], k = 0, \pm 1, \pm 2, \ldots$$

Example: Find square roots of $2i$

$2i = 2(\cos \pi/2 + i \sin \pi/2)$

$z_0 = 2e^{i\pi/2}$

$Z = \sqrt{2}\, exp\left[i\left(\frac{\pi/2}{n} + \frac{2k\pi}{2}\right)\right] k \in \mathbb{Z}$

$k = 0$

$z_1 = 1 + i$, $z_2 = -1 - i$

$= \theta_0 + 2k\pi, k \in \mathbb{Z}$

$= \frac{\theta_0}{n} + \frac{2k\pi}{n}, k \in \mathbb{Z}$

$exp\left[i\frac{\theta_0}{n} + \frac{2k\pi}{n}\right], k \in \mathbb{Z}$

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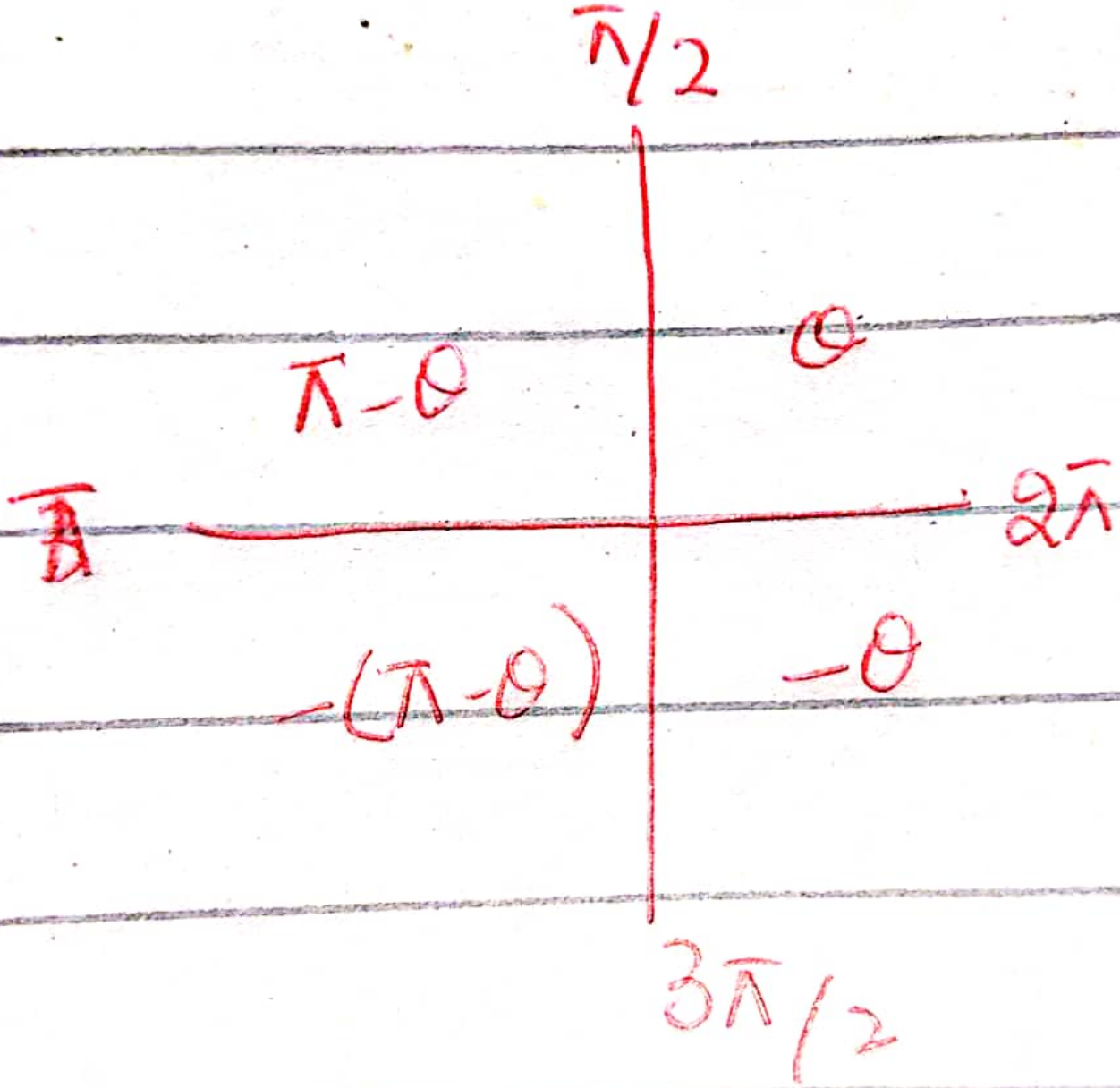
Polynomial Functions

Example

$$f(z) = \frac{1}{z} = z^{-1}$$

is NOT a polynomial.

Power of function in polynomial is non-negative and finite



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Relation between Real & Complex Limit

Example: Compute the limit

$$\lim_{z \to i} z^2 + z + 1 .$$

Solution:

$f(z) = z^2 + z + 1$

$f(x+iy) = (x+iy)^2 + (x+iy) + 1$

$= (x^2 - y^2) - 2ixy + (x+iy) + 1$

$= \underbrace{(x^2 - y^2 + x + 1)}_{u(x,y)} + i\underbrace{(-2xy + y)}_{v(x,y)}$

$\lim_{(x,y) \to (0,1)} x^2 - y^2 + x + 1 = -1 + 1 = 0$

$\lim_{(x,y) \to (0,1)} -2xy + y = 1$

$\Rightarrow \lim_{z \to i} f(z) = i$

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$(2i)^{1/2}$ $\quad 2i$

$2i = r$

$r = 2$

$\theta = \pi/2$

Now

$2i = 2e^{i\pi/2}$

$(2i)^{1/2} = (2e^{i\pi/2})^{1/2}$

$= \sqrt{2}\, e^{i(\pi/4) + \frac{2k\pi}{2}}$

$k = 0$

$\sqrt{2}\, e^{i\pi/4}$

$\sqrt{2}(\cos\pi/4 + i\sin\pi/4)$

$\sqrt{2}\left(\frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}\right)$

$z_1 = 1 + i$

$-\sqrt{2}(\cos\pi/4 - i\sin\pi/4)$

$z_2 = -1 - i$

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an angle)

Q1 Find Square roots of $2i$

$2i = 2(\cos\frac{\pi}{2} + i\sin\pi/2)$

$z_0 = 2e^{i\pi/2}$

$z = \exp[i(\frac{\theta_0}{n} + \frac{2k\pi}{n})]$

$z = \sqrt{2}[$

Let $z = \sqrt{2i}$

by using exponential form $z = re^{i\theta}$

$\Rightarrow r^2 e^{i(2\theta)} = 2i = 2e^{i(\pi/2 + 2n\pi)}$

$r^2 = 2 \Rightarrow r = \sqrt{2}$

$2\theta = \frac{\pi}{2} + 2\pi\pi \Rightarrow \theta = \frac{\pi}{4} + n\pi$

So, $z = \sqrt{2}e^{i(\pi/4 + 2n\pi)}$

$z = \sqrt{2}\left[\cos(\frac{\pi}{4} + 2n\pi) + i\sin(\frac{\pi}{4} + 2n\pi)\right]$

$z = \sqrt{2}\left[\pm\frac{1}{\sqrt{2}} \pm \frac{1}{\sqrt{2}}i\right]$

$z = \pm 1 \pm i$

$z = 1+i$, $z = -1-i$

Powers in Exponen

onential form

er $z = re^{i\theta}$

$e^{2\pi i}$ for $n = 0, \pm 1, \pm 2$

tical induction)

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Derivatives

Example: If $f(z) = z^2$ then show that $f'(z) = 2z$.

Solution;

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(z+\Delta z)^2 - z^2}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{z^2 + 2z\Delta z + \Delta z^2 - z^2}{\Delta z}$$

$$= \lim_{\Delta z \to 0} \frac{(2z + \Delta z)\Delta z}{\Delta z} = \lim_{\Delta z \to 0} (2z + \Delta z) = 2z$$

VU

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Properties of Complex Limits

Theorem: Suppose that $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$

Then

1. $\lim_{z \to z_0} c\,f(z) = cA$, c a complex constant.

$f(z) = u(x,y) + i\,v(x,y)$, $A = u_0 + iv_0$

$\Rightarrow \lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$ & $\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$

$$cA = (c_0 + ic_1)(u_0 + iv_0) = (c_0u_0 - c_1v_0) + i(c_1u_0 + v_0c_0)$$

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Consequences of Continuity

Theorem: If a function f is continuous throughout a region R that is both closed and bounded, there exists a nonnegative real number M such that

$$|f(z)| \leq M \text{ for all points } z \text{ in } R$$

where equality holds for at least one such z.

Proof: Given f is continuous. We know that $g(z) = |z|$ is continuous $g \circ f(z) = g(f(z)) = |f(z)|$

Since composition of continuous functions is continuous so $|f(z)|$ is continuous. If $f(z) = u(x,y) + iv(x,y)$ then $\sqrt{(u(x,y))^2 + (v(x,y))^2}$ is continuous.



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$$z = \pm 1 \pm i$$

$z = 1+i$, $z = -1-i$

Q2: formula of nth root of Complex number

$$z_0^{1/n} \; z = \sqrt[n]{r_0}\exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right]$$

$k = 0, \pm 1, \pm 2, \cdots$

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View Help

Complex Limits

Example: Show that

$$\lim_{z \to z_0} \bar{z} = \bar{z}_0$$

Solution:

$\varepsilon > 0$ ✓

$|f(z) - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\bar{z} - \bar{z}_0| < \varepsilon$

$\Leftrightarrow |\overline{z - z_0}| < \varepsilon$

$\Leftrightarrow |z - z_0| < \varepsilon$ ✓

$\delta = \varepsilon$

$|z - z_0| < \delta = \varepsilon$

$\Rightarrow |f(z) - \bar{z}_0| < \varepsilon$

$|f(z) - \bar{z}_0| < \varepsilon$ whenever $|z - z_0| < \delta$

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oots of Compl...

Relation between roots of a complex number and roots of unity

ample: Find all values of $(2i)^{1/3}$.

step 1: $z_0 = 2i = 2\cos(\pi/2) + i\sin\pi/2$

$z_0 = 2e^{i(\pi/2)}$

$z_0 = 2e^{i(\pi/2 + 2k\pi)} \quad k \in \mathbb{Z}$

2: $z_0^{1/3} = \sqrt[3]{2}\, e^{i\left(\frac{\pi}{6} + \frac{2k\pi}{3}\right)} \quad k \in \mathbb{Z}$

3: $k = 0, 1, 2$

$c_0 = \sqrt[3]{2}\, e^{i(\pi/6)} = \sqrt[3]{2}\left(\frac{\sqrt{3}}{2} + i/2\right)$

$c_1 = c_0 w_3, \quad c_2 = c_0 w_3^2$

Quadratic

Consider equation

Then the quadratic form... the above equation

ax^2

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nuity of Complex Functions

Show that the function $f(z) = \bar{z}$ is continuous.

$\lim_{z \to z_0} \bar{z} = \bar{z_0}$ (From previous discussion)

$f(z_0) = \bar{z_0}$

$\Rightarrow$ $f(z)$ is Cont. at z_0

$\Rightarrow$ $f(z)$ is Cont. at each Complex number.

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$$\lim_{z \to 4i} \frac{z^2 + 16}{z - 4i}$$

$$\frac{z^2 - (-16)}{z^2 - 4i^2}$$

$$\frac{z^2 - 4i^2}{z - 4i}$$

$$\frac{(z - 4i)(z + 4i)}{(z - 4i)}$$

$$8i$$

Q1 Re z and Im z

$z = x + iy \qquad \bar{z} = x - iy$

$z + \bar{z} = x + iy + x - iy$

$z + \bar{z} = 2x$

$2x = z + \bar{z}$

$x = \frac{z + \bar{z}}{2}$

Im z $\quad z - \bar{z} = x + iy - x + iy$

$z - \bar{z} = +2iy$

$y = \frac{z - \bar{z}}{2i}$

$Re\, z = \frac{z + \bar{z}}{2} \qquad Im\, z = \frac{z - \bar{z}}{2i}$

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$$\lim_{z \to \infty} \frac{z^2}{z^2 + z - 1}$$

$$= \lim_{z \to \infty} \frac{z^2\left(1 + \frac{2}{z}\right)}{z^2\left(1 + \frac{1}{z} - \frac{1}{z^2}\right)}$$

$$= \frac{\left(1 + \frac{2}{\infty}\right)}{1 + \left(\frac{1}{\infty}\right) - \left(\frac{1}{\infty}\right)}$$

$$= 1$$

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c) $(1-i)^4 = ?$

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1 - 1 - 2i = 2i$$
$$(1-i)^4 = (1-i)^2 (1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

2. Show that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

3. use the associative and commutative laws for multiplication to

$$L.H.S = (z_1 z_2)(z_3 z_4)$$
$$= (z_2 z_1)(z_3 z_4) \qquad \because (z_1 z_2) = (z_2 z_1) \quad \textit{commutative la}$$
$$= (z_2)(z_1 z_3)(z_4) \qquad \because (z_1)(z_3 z_4) = (z_1 z_3)(z_4) \quad \textit{associati}$$
$$= (z_1 z_3)(z_2)(z_4) \qquad \because (z_2)(z_1 z_3) = (z_1 z_3)(z_2) \quad \textit{commut}$$
$$= (z_1 z_3)(z_2 z_4) \qquad \because (z_2)(z_4) = (z_2 z_4)$$
$$= R.H.S$$

5. Drive expression

$$\frac{z_1}{z_2} = \frac{x_1 x_2 - y_1 y_2}{x_2^2 + y_2^2} + i\frac{y_1 x_2 - x_1 y_2}{x_2^2 + y_2^2} \quad (z_2 \neq 0)$$

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ where $(z_2 \neq 0)$

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is given by $|z - z_0| = \varepsilon$

defined by $|z - z_0| \leq \varepsilon$ ✓

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operties of Complex Limits

e: Compute the limit

$$\lim_{z \to i} z^2 + z + 1 .$$

$$= \left(\lim_{z \to i} z\right)\left(\lim_{z \to i} z\right) + \lim_{z \to i} z + \lim_{z \to i} 1$$

$$= (i)(i) + i + 1$$

$$= -1 + i + 1 = i$$

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You
Yesterday, 6:44 pm

(i) Square root of $\sqrt{(-3)^2}$ = ?

(ii) Limit $\lim_{z \to i} \frac{iz^3 - 1}{z + i}$

(iii) Square of $(-2 - 5i)$

(iv) $f(z) = \frac{1}{z^2 + 4}$ is discontinuous at point.

(v) $f((z)) = z^3 - 3z + 12i + 5$

$f'(z)$ = ?

(vi) if $z = 3 + i4$ and $|z| = 5$

Then $|\bar{z}|$ = ?

(vii) if $w = 1 + 7i$ and

$z = 3 + 2i$

Then find $z \cdot w$ = ?

~~Square root~~

(viii) Simplify. $\left[\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right]^{12}$

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Date

$$\lim_{z \to i} \frac{z - i}{z^2 + 1}$$

$$\lim_{z \to i} \frac{z - i}{z^2 - (-1)} \qquad -1 = i^2$$

$$\frac{z - i}{z^2 - i^2}$$

~~$z = i$~~

$$\frac{}{(z - i)(z + i)}$$

$$\lim_{z \to i} \frac{1}{z + i}$$

$$\frac{1}{2i}$$

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$$\lim_{z \to i} \frac{z^3}{z^2 + i}$$

Date

$$\frac{i^3}{i^2 + i}$$

$$\frac{i^2(i^2)}{i^2(i+1)}$$

$$\frac{-1}{1+i}$$

$$= \frac{-1}{1+i} \times \frac{1-i}{1-i}$$

$$\frac{-1(1-i)}{1+1} \qquad \frac{-1}{2}$$

$$\frac{-1(1-i)}{2} = -\frac{1}{2} + \frac{1}{2}i$$

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$f(z) = z^2$ $\quad f'(z) = 2z$

$z = x + iy$ $\quad f(z) = z^2$

$z^2 = (x+iy)^2 = x^2 - y^2 + i2xy$

$u(x,y) = x^2 - y^2$ $\quad v(x,y) = 2xy$

$u_x = 2x$ $\quad v_y = 2x$

$u_y = -2y$ $\quad v_x = 2y$

$u_x = v_y$ $\quad u_y = -v_x$

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$$f(z) = (z - 4i)^8 \quad , z = 3 + 4i$$

$$f'(z) = 8(z - 4i)^7 (z - 4i)' =$$

$$f'(3+4i) = 8(3 + 4i - 4i)^7$$

$$= 8 \cdot 3^7$$

$$= 17496$$

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Product and Powers in Exponential Form

Product in exponential form

For complex number $z = re^{i\theta}$

$z^n = r^n e^{in\theta}$ for $n = 0, \pm 1, \pm 2, \pm 3, \ldots$

Example:

$$z = 7e^{i\frac{\pi}{4}}$$

$$z^{100} = 7^{100} e^{i\frac{\pi}{4}(100)}$$

$$z^{1000} = 7^{1000} e^{i\frac{\pi}{4}(1000)}$$

Product and Powers in Exponential Form

Example: Calculate $(\sqrt{3}+i)^3$ in exponential form

$r = \sqrt{3+1} = 2$

$\theta = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$

$$\sqrt{3}+i = 2e^{i\frac{\pi}{6}}$$

$$(\sqrt{3}+i)^3 = 2^3 e^{i\frac{\pi}{6}(3)}$$

$$= 8e^{i\frac{\pi}{2}}$$ Ans

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differenciate.

(iv) $(\sqrt{3} - i)^{1/3}$, exponencial form.

(5 Marks)

$1,3$

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MTH632 (Solved-Excercised).pdf...

$(1-i)^4$ =?

$$(1-i)^2 = (1)^2 + (i)^2 - 2(1)(i)$$
$$= 1 - 1 - 2i = 2i$$
$$(1-i)^4 = (1-i)^2(1-i)^2$$
$$= (2i)(2i)$$
$$= 4i^2 = 4(-1) = -4$$

how that

$$L.H.S = \frac{1}{1/z} = \frac{1}{z^{-1}} \qquad \because \frac{1}{z} = z^{-1}$$
$$= \frac{1}{z^{-1}} \cdot \frac{z}{z} = \frac{1.z}{z^{-1}.z}$$
$$= \frac{z}{1} = z = R.H.S$$

use the associative and commutative laws for m

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$$-8i = 8\exp\left[i\left(\frac{-\pi}{2} + 2k\pi\right)\right].$$

$k = 2$ find $(8i)^{1/3}$.

$$(-8i)^{1/3} = \left(8\exp\left[i\left(\frac{-\pi}{2} + 2k\pi\right)\right]\right)^{1/3}.$$

$$= 8^{1/3}\exp\left[i\left(\frac{-\pi}{6} + \frac{2k\pi}{3}\right)\right].$$

$k = 2$.

$$= 2^{3\times 1/3}\exp\left[i\left(\frac{-\pi}{6} + \frac{4\pi}{3}\right)\right].$$

$$= 2\exp\left[i\left(\frac{-\pi + 8\pi}{6}\right)\right].$$

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(i) $\dfrac{(\cos 2\theta + i\sin 2\theta)^5 (\cos 3\theta - i\sin 3\theta)^6}{(\cos 4\theta - i\sin 4\theta)^7 (\cos 5\theta + i\sin 5\theta)^8}$

$= \dfrac{(\cos 2\theta + i\sin 2\theta)^5 (\cos(-3\theta) + i\sin(-3\theta))^6}{(\cos(-4\theta) + i\sin(-4\theta))^7 (\cos 5\theta + i\sin 5\theta)^8}$

$= \dfrac{(\cos 10\theta + i\sin 10\theta)(\cos(-18\theta) + i\sin(-18\theta))}{(\cos(-28\theta) + i\sin(-28\theta))(\cos 40\theta + i\sin 40\theta)}$ by De. Moivre's theorem

$= \dfrac{\cos(10\theta - 18\theta) + i\sin(10\theta - 18\theta)}{\cos(-28\theta + 40\theta) + i\sin(-28\theta + 40\theta)}$

$= \dfrac{\cos(-8\theta) + i\sin(-8\theta)}{\cos 12\theta + i\sin 12\theta}$

$= \cos(-8\theta - 12\theta) + i\sin(-8\theta - 12\theta) = \cos(-20)\theta + i\sin(-20)\theta$

$= \cos 20\theta - i\sin 20\theta$

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$$\int_0^1 (x - x^3)\, dx$$

$$\left| \frac{x^2}{2} - \frac{x^4}{4} \right|_0^1$$

$$= \left[\frac{(1)^2}{2} - \frac{(1)^4}{4} \right] - \left[\frac{(0)^2}{2} - \frac{(0)^4}{4} \right]$$

$$= \frac{1}{2} - \frac{1}{4}$$

$$= \frac{2-1}{4} = \frac{1}{4} \text{ Ans.}$$

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Derivative of:

$f(z) = \frac{z^3}{z+i}$ at $z = i$

$$f'(z) = \frac{(z+i)3z^2 - z^3(1)}{(z+i)^2}$$

$$= \frac{3z^2(z+i) - z^3}{(z+i)^2}$$

$$f'(i) = \frac{3(i^2)(i+i) - (i^3)}{(i+i)^2}$$

$$= \frac{3(-1)(2i) - (-i)}{(2i)^2}$$

$$= \frac{-6i + i^2}{-4} \Rightarrow \frac{3}{2}i + \frac{1}{4}$$

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(5) Evaluate

$\frac{1}{2+i} + 2$

$\Rightarrow \frac{1+i}{2+i} + 2$

(6) verify

$(3,1)(3,-1)\left\{\frac{1}{5}, \frac{1}{10}\right) = (2,1)$

$(3+i)(3-i)\left\{\frac{1}{5} + \frac{1}{10}i\right)$

$(9-i^2)\left(\frac{1}{10} + \frac{1}{10}i\right)$

$(10)\left(\frac{1}{5} + \frac{1}{10}i\right)$

$10 \times \frac{1}{5} + 10 \times \frac{1}{10}i$

$(2+i)$ Nayab Paper Products $(2,1)$



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$$\left(\sqrt{3}-i\right)^{\frac{1}{3}}$$

Solution;

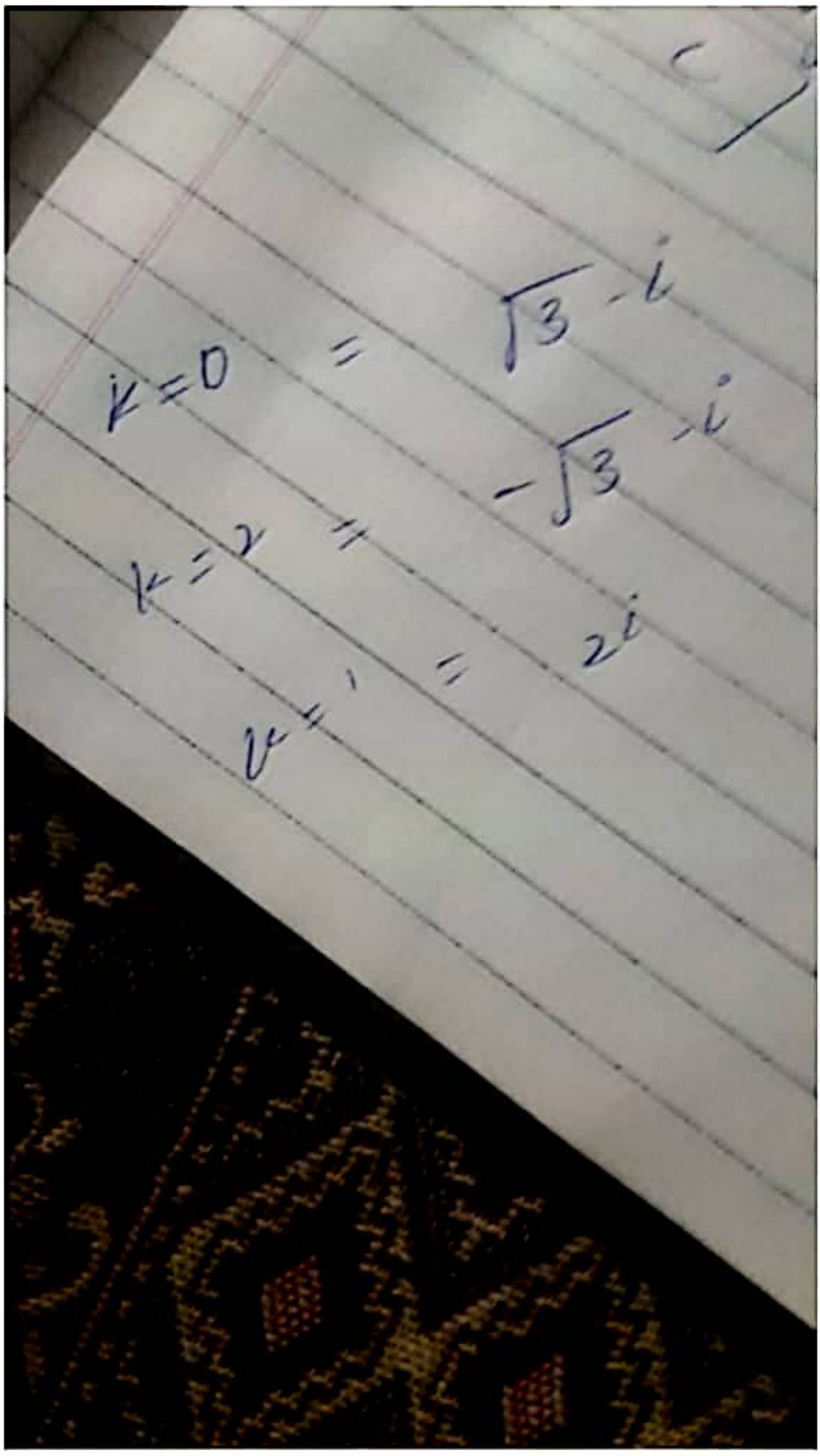
$$\sqrt{3}-i=re^{i\theta}$$

$$r=2$$

$$\theta=-\frac{\pi}{6}$$

$$\left(\sqrt{3}-i\right)^{\frac{1}{3}}=\sqrt[3]{2}\ e^{-\frac{i\pi}{18}}\quad ans$$

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e form

$L(z) = Az + B$

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$$g(f(z)) = \frac{1}{z^3} = \frac{1}{-i}$$

$$= \frac{1}{-i} \times \frac{i}{i}$$

$$= i$$

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$$a_n = 2^n$$

$$a_n = (-1)^n$$

$$a_n = -n$$

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Domain of the Functions

(36) $f(z) = \frac{1}{z^2+1}$, Ans $\boxed{z \neq \pm i}$

(37) $f(z) = \frac{1}{z^2-1}$, Ans $\boxed{z \neq \pm 1}$

(38) $f(z) = \frac{1}{z^2}$, Ans $\boxed{z = 0}$

(39) $f(z) = \frac{1}{z+\bar{z}}$, Ans $\boxed{z \neq \pm i}$

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$$U_y = 1$$

and

$$U_y = -V_x$$

$$1 = -V_x$$

$$\boxed{V_x = -1}$$

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Section 10

Examples:

1. Find the square root of 2i, Topic 12 time 15 mint

Answer:

We can write square root of 2i as $(2i)^{\frac{1}{2}}$

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right) = 2\exp\left[i\left(\frac{\pi}{2} + 2k\pi\right)\right] \qquad k=0,1$$

$$=2e^{i\left(\frac{\pi}{2}+2k\pi\right)}$$

Taking square root

$$(2i)^{\frac{1}{2}} = \left(2e^{i\left(\frac{\pi}{2}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(\frac{\pi}{4}+k\pi\right)}$$

When k=0

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+0\pi\right)} = \sqrt{2}e^{i\left(\frac{\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 1+i$$

When k=1

$$\sqrt{2}e^{i\left(\frac{\pi}{4}+1\pi\right)} = \sqrt{2}e^{i\left(\frac{5\pi}{4}\right)} = \sqrt{2}\left(\cos\frac{5\pi}{4} + i\sin\frac{5\pi}{4}\right) = -1-i = -(1+i)$$

So the roots of 2i are $\pm(1+i)$

1. Find the square root of $1-\sqrt{3}i$, Topic 12 time 15 mint

Solution:

$$1-\sqrt{3}i = \cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right) = 2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}$$

Taking square root

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \left(2e^{i\left(-\frac{\pi}{6}+2k\pi\right)}\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+k\pi\right)} \qquad k=0,1$$

When k=0

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+0\pi\right)}$$

$$= \sqrt{2}e^{i\left(-\frac{\pi}{12}\right)} = \sqrt{2}\left(\cos\left(-\frac{\pi}{12}\right) + i\sin\left(-\frac{\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\cos\left(\frac{\pi}{12}\right) - i\sin\left(\frac{\pi}{12}\right)\right) = -\sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = -\frac{\sqrt{3}-i}{\sqrt{2}}$$

When k=1

$$\left(1-\sqrt{3}i\right)^{\frac{1}{2}} = \sqrt{2}e^{i\left(-\frac{\pi}{12}+1\pi\right)}$$

$$= \sqrt{2}e^{i\left(\frac{11\pi}{12}\right)} = \sqrt{2}\left(\cos\left(\frac{11\pi}{12}\right) + i\sin\left(\frac{11\pi}{12}\right)\right)$$

$$= \sqrt{2}\left(\frac{\sqrt{3}-i}{2}\right) = +\frac{\sqrt{3}-i}{\sqrt{2}}$$

So the roots of $1-\sqrt{3}i$ are $\pm\frac{\sqrt{3}-i}{\sqrt{2}}$

2. $(-16)^{\frac{1}{4}} = (-1*16)^{\frac{1}{4}} = \left(i^2 \times 2^4\right)^{\frac{1}{4}} = 2(i)^{\frac{1}{2}} = 2\sqrt{i}$

We find the square root of i.

Topic 25 examples:

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4. show that $|e^{i\theta}-1|=2$

Solution:

$$
\begin{aligned}
&LHS=|e^{i\theta}-1| \\
&=|\cos\theta+i\sin\theta-1| \qquad 0\le\theta\le 2\pi \\
&when\ \ \theta=\pi \\
&=|\cos\pi+i\sin\pi-1| \\
&=|-1+0-1| \\
&=|-2|=2=RHS
\end{aligned}
$$

5. By writing the individual factors on the left in exponential form, performing the needed operations and changing back to rectangular coordinates, show that

a): $i(1-\sqrt{3}i)(\sqrt{3}+i)=2(1+\sqrt{3}i)$

Solution:

Exponential form are



$$
\begin{aligned}
&i=e^{i\frac{\pi}{2}}=\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}=i \\
&(1-\sqrt{3}i)=2\cos\frac{\pi}{3}-2i\sin\frac{\pi}{3}=2e^{-i\frac{\pi}{3}} \\
&(\sqrt{3}+i)=2\cos\frac{\pi}{6}+2i\sin\frac{\pi}{6}=2e^{i\frac{\pi}{6}}
\end{aligned}
$$

$$
\begin{aligned}
&LHS=i(1-\sqrt{3}i)(\sqrt{3}+i) \\
&=e^{i\frac{\pi}{2}}\left(2e^{-i\frac{\pi}{3}}\right)\left(2e^{i\frac{\pi}{6}}\right) \\
&=4e^{i\frac{\pi}{2}-\frac{\pi}{3}+\frac{\pi}{6}} \\
&=4e^{i\frac{3\pi-2\pi+\pi}{6}} \\
&=4e^{i\frac{\pi}{3}} \\
&=2(2e^{i\frac{\pi}{3}}) \\
&=2(1+\sqrt{3}i) \\
&=RHS
\end{aligned}
$$

b): $LHS=\dfrac{5i}{2+i}=\dfrac{5e^{i\frac{\pi}{2}}}{\sqrt{5}e^{i}}$

8. Prove: $\exp\left(i\dfrac{\theta_1+\theta_2}{2}\right)\exp\left(i\dfrac{\theta_1-\theta_2}{2}\right)=\exp(i\theta_1)$

Solution:

$$
\begin{aligned}
&LHS=\exp\left(i\frac{\theta_1+\theta_2}{2}\right)\exp\left(i\frac{\theta_1-\theta_2}{2}\right) \\
&=\left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}\right)\right)\left(\cos\left(\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right) \\
&=\left(\cos\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)-\sin\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right) \\
&\quad+i\left(\sin\left(\frac{\theta_1+\theta_2}{2}\right)\cos\left(\frac{\theta_1-\theta_2}{2}\right)+\cos\left(\frac{\theta_1+\theta_2}{2}\right)\sin\left(\frac{\theta_1-\theta_2}{2}\right)\right) \\
&=\cos\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right)+i\sin\left(\frac{\theta_1+\theta_2}{2}+\frac{\theta_1-\theta_2}{2}\right) \\
&=\cos(\theta_1)+i\sin(\theta_1) \\
&=e^{i\theta_1}=\exp(i\theta_1)=RHS
\end{aligned}
$$

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$$\frac{z_1}{z_2}=\frac{8+i}{3+8i}$$

$$=\frac{8+i}{3+8i}\times\frac{3-8i}{3-8i}$$

$$=\frac{24+64i-3i-8i^2}{(3)^2-(8i)^2}$$

$$=\frac{24+64i-3i-8(-1)}{9-64(-1)}$$

$$=\frac{24+64i-3i+8}{9-64(-1)}$$

$$=\frac{32+61i}{73}$$

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Q5: Simplify $(\sqrt{3}+i\,3)-i(5-\sqrt{3}i)$

Solution:

$$(\sqrt{3}+i\,3)-i(5-\sqrt{3}i)$$

$$=\sqrt{3}+3i-5i+\sqrt{3}i^2$$

$$=\sqrt{3}-2i+\sqrt{3}(-1)$$

$$=\sqrt{3}-2i-\sqrt{3}$$

$$=\sqrt{3}-\sqrt{3}-2i$$

$$=0-2i$$

$$=-2i$$

Q6: Find *Re(z)* and *Im(z)* complex numbers, $z = -3 + 9i$.

Solution:

$Re(z) = -3$ and $Im(z)=9$

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MTH632 (Solved-Excerc...

SOLVED EXERCISE MTH-632

Complex Analysis & Differential Geometry

Instructor

Dr. Sohail Iqbal

Midterm's Syllabus Solved Exercises

Book Name: Brown-Churchill-Complex Variables and Application 8th edition

Chapter 1 Section 2,3,4,8,10,

Chapter 2 Section 12,18,20,23,25,26

Section 2 ***Basic Algebraic Operations***

Question No.1: **Verify that**

a): $(\sqrt{2}-i)-i(1-\sqrt{2})=-2i$

proof:

$$\begin{aligned} L.H.S &= (\sqrt{2}-i)-i(1-\sqrt{2}) \\ &= \sqrt{2}-i-i+\sqrt{2}i^2 \\ &= \sqrt{2}-2i+\sqrt{2}(-1) \\ &= \sqrt{2}-2i-\sqrt{2} \\ &= -2i = R.HS \end{aligned}$$

b):

$$\begin{aligned} L.H.S &= (2,-3)(-2,1) \\ &= (2-3i)(-2+i) \\ &= 2(-2+i)-3i(-2+i) \\ &= -4+2i+6i-3i^2 \\ &= -4+8i-3(-1) \\ &= -4+8i+3 \\ &= -1+8i = R.HS \end{aligned}$$

c):

$$\begin{aligned} L.H.S &= (3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right) \\ &= (3+i)(3-i)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= \left(3(3-i)+i(3-i)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= \left(9-3i+3i-i^2\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= \left(9-(-1)\right)\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= 10\left(\frac{1}{5}+\frac{1}{10}i\right) \\ &= 2+i \\ &= (2,1) = R.HS \end{aligned}$$

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Example: The Polar representation of z = -1-i is given by:

$$z = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$$

or $\frac{\pi}{4}$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Exponential Form of z is:

$$-i\left(-\frac{3\pi}{4}\right) \qquad \sqrt{2}\,e^{-i\left(\frac{3\pi}{4}\right)}$$

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$$-\pi < \theta \leq \pi$$

Example: Calculate Polar representation of $z = -1 - i$

$$r = |z| = \sqrt{(-1)2 + (-1)2} = \sqrt{1+1} = \sqrt{2}$$

$$\tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-1}{-1}\right) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$\theta = \arg z = \frac{\pi}{4} + \pi = \frac{5\pi}{4}$$

$$-\pi < \theta \leq \pi \qquad \text{Arg } z = \frac{5\pi}{4}$$

Polar representation of $\mathbf{z = \sqrt{2}\left(cos\frac{5\pi}{4} + isin\frac{5\pi}{4}\right)}$

If x = 0 then we can not use the following to find arg z

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find $f'(z) = ?$

$$f(z) = \frac{(1+z^2)^4}{z^2}, \quad z \neq 0$$

$$f'(z) = \frac{4(1+z^2)^3 2z}{2z}$$

$$f'(z) = 4(1+z^2)^3 \quad \text{Ans.}$$

$$\lim_{z \to i} \frac{z-i}{z^2+1}$$

$$= \lim_{z \to i} \frac{z-i}{z^2+1}$$

$$= \lim_{z \to i} \frac{z-i}{z^2-(-1)}$$

$$= \lim_{z \to i} \frac{z-i}{z^2-(i)^2}$$

$$= \lim_{z \to i} \frac{z-i}{(z+i)(z-i)}$$

$$= \lim_{z \to i} \frac{1}{z+i}$$

$$= \frac{1}{i+i} = \frac{1}{2i} \quad \text{Ans.}$$

$f'(z) = ?$ at $z = i$

$$f(z) = \frac{z^3}{(z+i)^3}$$

$$f'(z) = \frac{3z^2}{3(z+i)^2(1)}$$

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$$f'(z) = \frac{3z^2}{3(z+i)^2}$$

$$f'(i) = \frac{3(i)^2}{3(i+i)^2}$$

$$= \frac{3(-1)}{3(2i)^2}$$

$$= \frac{-3}{3 \cdot 4i^2}$$

$$= \frac{-1}{4i^2}$$

$$= \frac{-1}{4(-1)}$$

$$= \frac{+1}{+4} = \frac{1}{4}$$

Find derivative at $z = 2i$ of

$$f(z) = (iz^3 + 3z^2)^3$$

$$f(z) = (iz^3 + 3z^2)^3$$

$$f'(z) = \frac{d}{dz}(iz^3 + 3z^2)^3$$

$$= 3(iz^3 + 3z^2)^2 (3z^2 i + 6z)$$

at $z = 2i$

$$= 3\left[i(2i)^3 + 3(2i)^2\right]^2 \left[3(2i)^2 i + 6(2i)\right]$$

$$= 3\left[i(8i^3) + 3(4i^2)\right]^2 (12i^3 + 12i)$$

$$= 3(8i^4 + 12i^2)^2 (12i^3 + 12i)$$

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$= 3(8-12)^2(-12i+12i)$

$= 3(16)\quad(0)$

$= 0$

(5) Evaluate $\lim_{z \to 4i} \frac{z^2+16}{z-4i}$

$\lim_{z \to 4i} \frac{z^2-(-16)}{z-4i}$

$\lim_{z \to 4i} \frac{z^2-16i}{z-4i}$

$\lim_{z \to 4i} \frac{(z)^2-(4i)^2}{z-4i}$

$= \lim_{z \to 4i} \frac{(z-4i)(z+4i)}{(z-4i)}$

$= \lim_{z \to 4i} (z+4i)$

$= 4i+4i = 8i$

$a^2-b^2 = (a+b)(a-b)$

$i^2 = -1$

(6) Evaluate $(-4-4i)^3$

$z = -4-4i$

$r = \sqrt{(4)^2+(4)^2}$

$= \sqrt{16+16}$

$= \sqrt{32}$

$= 4\sqrt{2}$

both are +ve

$\theta = \tan^{-1}(y/x)$

$= \tan^{-1}(1)$

$= \frac{\pi}{4}$

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(4)

$\frac{\pi + \frac{\pi}{4}}{1} = \frac{4\pi + \pi}{4} = \frac{5\pi}{4}$

$\text{Arg}(z) = \theta = \pi + \theta$

$= \pi + \frac{\pi}{4} = \frac{5\pi}{4}$

$z = -4 - 4i = (4\sqrt{2}\, e^{i 5\pi/4})$

$(-4-4i)^3 = (4\sqrt{2}\, e^{i5\pi/4})^3$

2. Prove $\lim_{z \to \infty} \frac{4z^2}{(z-1)^2} = 4$

$\frac{4z^2}{z^2(1-$

$= \lim_{z \to \infty} \frac{4z^2}{z^2 (1 - \frac{1}{z})^2}$

$= \lim_{z \to \infty} \frac{4}{(1 - \frac{1}{z})^2} = \frac{4}{(1 - \frac{1}{\infty})^2} = \frac{4}{(1-0$

Simpfy by using De-Moivre's formula.

$$\frac{(\cos 2\theta + i \sin 2\theta)^5 \, (\cos 3\theta - i \sin 3\theta)^6}{(\cos 4\theta - i \sin 4\theta)^7 \, (\cos 5\theta + i \sin 5\theta)^8}$$

by using De-moivre's formula.

$= \frac{(e^{i2\theta})^5 (e^{-i3\theta})^6}{(e^{-i4\theta})^7 (e^{i5\theta})^8} = \frac{(e^{i\theta})^{10-18}}{(e^{i\theta})^{-28+}}$

$= (e^{i\theta})^{-8-12} \Rightarrow (e^{i\theta})^{-20}$

$= (\cos 20\theta - i \sin 20\theta)$ Ans.

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(9) Find derivative $f(z) = (1+4z)^3$

$f'(z) = 3(1+4z)^2 (4)$

$= 12(1 + 16z^2 + 8z)$

$= 12 + 192z^2 + 96z$.

(10) $f(z) = z^4 - 9z^2 + iz - 2$

check continuity at $z = i$

$f(z) = z^4 - 9z^2 + iz - 2$

$= (i)^4 - 9(i)^2 + (i)(i) - 2$

$= (i^2)(i)^2 - 9(-1) + (i^2) - 2$

$= (-1)(-1) + 9 + (-1) - 2$

$= 1 + 9 - 1 - 2 \Rightarrow 9 - 2$

$= 7$

(11) $(\sqrt{3} - i)^{1/3}$ exponential form

Solve:-

in exponential form $r = \sqrt{3+1} = 2$

$\theta = \tan^{-1}\frac{1}{\sqrt{3}} = \frac{\pi}{6}$

$z = e^{i\theta}$

$\sqrt{3} - i = 2e^{i\frac{\pi}{6}}$

$(\sqrt{3} - i)^3 = 2^3 e^{i\frac{\pi}{6}(3)}$

$= 8e^{i\frac{\pi}{2}}$

(12) Verify That:-

$(\sqrt{2} - i) - i(1 - \sqrt{2}i) = -2i$

L.H.S $= (\sqrt{2} - i) - i(1 - \sqrt{2}i)$

$= \sqrt{2} - i - i + \sqrt{2}i^2$

$= \sqrt{2} - 2i + \sqrt{2}(-1)$

$= \sqrt{2} - 2i - \sqrt{2}$

$= -2i$ Ans.

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(13) If $w = 1+7i$ and $z = 3+2i$
Then find $z \cdot w = ?$

Solve:-

$$z \cdot w = (3+2i)(1+7i)$$
$$= 3 + 21i + 2i + 14i^2$$
$$= 3 + 23i + 14(-1)$$
$$= 3 + 23i - 14$$
$$= -11 + 23i$$

(14) $z_1 = \frac{1}{e^{i\pi/3}}$, $z_2 = \frac{1}{e^{i\pi/2}}$

$$\frac{z_1}{z_2} = ?$$

$$\frac{z_1}{z_2} = \frac{\frac{1}{e^{i\pi/3}}}{\frac{1}{e^{i\pi/2}}} = \frac{1}{e^{i\pi/3}} \times e^{i\pi/2}$$

$$= e^{i\pi/3} \cdot e^{i\pi/2}$$
$$= e^{i(\pi/3 \cdot \pi/2)}$$
$$= e^{i\pi/6}.$$

(15) $f(z) = (z + 3i + 5)^7$ $z = -3i + 5$

$f'(z) = ?$

$f'(z) = 7(z+3i+5)^6$

$z = -3i + 5$

$f'(z) = 7(-3i + 5 + 3i + 5)^6$

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$= 7(10)^6$

(16) magnitude of $\exp(-2+3i)$

let $z = -2+3i$

$|z| = \sqrt{(-2)^2+(3)^2}$

$= \sqrt{4+9}$

$= \sqrt{13}$.

(17) Verify

$(3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right) = (2,1)$

L.H.S $= (3,1)(3,-1)\left(\frac{1}{5},\frac{1}{10}\right)$

$= (3+i)(3-i)\left(\frac{1}{5}+\frac{1}{10}i\right)$

$= 9-3i+3i-i^2\left(\frac{1}{5}+\frac{1}{10}i\right)$

$= 9-(-1)\left(\frac{1}{5}+\frac{1}{10}i\right)$

$= 10\left(\frac{1}{5}+\frac{1}{10}i\right)$

$= 2+i$

$= (2,1)$ R.H.S

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